Stochastic Processes 2021/22 Semester 1

Tutorial 9

1. Consider Question 3 from Tutorial 8.

   (a) Find all stationary distributions for the Markov process $X(t)$ and for the embedded Markov chain X_n^*. Are they the same? Why?

   (b) Find the limiting distribution for the Markov process that starts from (i) $X(0) = 1$ and (ii) $X(0) = 2$.

2. A continuous time Markov process $X(t)$ on the state space $S = \{1, 2, 3, 4, 5, 6\}$ has transition rate matrix given by

$$Q = \begin{pmatrix} -2 & 1 & 0 & 1 & 0 & 0 \\ 1 & -4 & 1 & 1 & 0 & 1 \\ 0 & 0 & -4 & 0 & 4 & 0 \\ 0 & 0 & 0 & -2 & 0 & 2 \\ 0 & 0 & 4 & 0 & -4 & 0 \\ 0 & 0 & 0 & 2 & 0 & -2 \end{pmatrix}$$

   (a) Write down the transition matrix of the corresponding jumping chain X_n^* . Hence classify the states of the process (identifying also transience/recurrence) and find the absorption probabilities associated with each of the closed classes.

   (b) Find all stationary distributions for the Markov process $X(t)$ and for the embedded Markov chain X_n^*. Are they the same? Why / why not?

   (c) Find the limiting distribution for the Markov process that starts (i) from $X(0) = 2$; (ii) from $X(0) = 3$.

3. Consider a Galton-Watson branching process with a distribution of the number of offsprings

$$p_0 = p_1 = 1/2, \quad p_k = 0 \quad \text{for} \quad k \geq 2.$$

   Find the expected time to reach 0 if the initial state is
   (a) $X_0 = 1$; (b) $X_0 = 2$.

4. Consider a Galton-Watson branching process with a distribution of the number of offsprings

$$p_1 = 1/3, p_2 = 2/3, \quad p_k = 0 \text{ for all other } k.$$

Find the expected time
(a) to hit the set $\{2, 3, 4, \ldots\}$ if $X_0 = 1$;
(b) to hit the set $\{3, 4, 5, \ldots\}$ if $X_0 = 2$;
(c) to hit the set $\{4, 5, 6, \ldots\}$ if $X_0 = 2$.

5. Consider a Galton-Watson process with a distribution of the number of offsprings

$$p_0 > 0, p_1 > 0, p_2 > 0, \text{ and } p_k = 0, \text{ for } k \geq 3.$$

Find the extinction probability if
(a) $X_0 = 1$ and $p_0 = 1/2, p_1 = 1/4, p_2 = 1/4$;
(b) $X_0 = 1$ and $p_0 = p_1 = p_2 = 1/3$;
(c) $X_0 = 3$ and $p_0 = p_1 = p_2 = 1/3$;
(d) $X_0 = 1$ and $p_0 = p_1 = 1/4, p_2 = 1/2$;
(e) $X_0 = 4$ and $p_0 = p_1 = 1/4, p_2 = 1/2$;
(f) $X_0 = 1$ and $p_0 = p_1 = 1/5, p_2 = 3/5$;
(g) $X_0 = 5$ and $p_0 = p_1 = 1/5, p_2 = 3/5$.